VELOOP Rewards

Product Analytics, Data Readiness and Data Management Blueprint

Final Consolidated Report | Data Analyst Internship, Task 01

Data authenticity: VELOOP is a real product under development. The dataset supplied for this assignment is fictional; it was profiled and audited (Deliverables 12 and 13) and covers the Users entity only. Counts taken from it describe that file, not VELOOP. Nothing in this document is an actual VELOOP statistic. Every event, field, KPI and rule is **Proposed** or **To Be Verified**; nothing about VELOOP's product is Confirmed.

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1. Conclusion

The assignment asked for a reliable, traceable and reusable analytical foundation for a developing product, not impressive numbers. That foundation now exists on paper: 20 product areas mapped, 42 events designed, 128 fields defined across 10 entities, 54 KPIs with reproducible definitions, 39 quality rules, 56 dashboard visuals, a developer handoff with testable acceptance criteria, a practical two-phase pipeline, an access and privacy model, a change process, and a prioritised backlog of 43 items.
It is a **proposal, not a verified state**. No official VELOOP documentation or engineering confirmation was available, so 45 of 50 unknowns are To Be Verified, 3 are Proposed awaiting a decision, and 2 are Confirmed (both about the supplied file). 7 of the ten assessed areas are at readiness level R1 (designed on paper) and 3 at R2 (checked against the supplied dataset). The dataset was profiled and audited (Deliverables 12 and 13), but it covers the Users entity only, so relationship tests beyond Users, most KPI tests and 21 of 39 quality rules could not be run.
The most important conclusions are: keep activity, validation and reward as separate facts; make the wallet an append-only, idempotent ledger with unique transaction IDs; keep amounts per unit; make quality a pipeline stage; and start with a simple pipeline that grows only when a named trigger is met. The six Critical backlog items, mostly stable identifiers, idempotency, the wallet ledger design, the common payload and the Critical events, come first.

2. Scope, method and evidence discipline

The sources were the task brief and, later, the supplied simulated dataset file. A public search for 'VELOOP Rewards' returned unrelated products with similar names and no official page, so those results were not used. Every product fact is therefore a proposal. Each document carries a status per item: **Confirmed** (verified with evidence; so far only two facts about the supplied file), **Proposed** (recommended; a proposal awaits a decision) or **To Be Verified** (unknown). Anything not known is recorded in the Assumption Register rather than guessed. Priorities are Critical, High, Medium or Low, each with a reason.

3. Deliverables and where to find them

The brief's 21 deliverables are listed with the files that satisfy them. Most file numbers match the brief; the exceptions are noted in the status column and in the last row of the file index.

#	Deliverable	Required form	File(s)	Status
01	Executive Summary	2 to 4 pages	Executive_Summary.pdf	Delivered
02	Product Data Audit	Detailed report	02_Product_Data_Audit.pdf	Delivered
03	Data Coverage Matrix	Spreadsheet	03_Data_Coverage_Matrix.xlsx (+ .pdf)	Delivered
04	User Analytics Specification	Document or spreadsheet	04_User_Analytics_Specification.pdf (+ .xlsx)	Delivered
05	Earning and Reward Analytics Specification	Document	05_Earning_Reward_Analytics_Specification.pdf (+ .xlsx)	Delivered
06	Wallet and Withdrawal Analytics Specification	Document	06_Wallet_Withdrawal_Analytics_Specification.pdf (+ .xlsx)	Delivered
07	Referral and Engagement Analytics	Document	07_Referral_Engagement_Analytics.pdf (+ .xlsx)	Delivered
08	Event Tracking Plan	Spreadsheet	08_Event_Tracking_Plan.xlsx (+ .pdf)	Delivered
09	Data Dictionary	Spreadsheet	09_Data_Dictionary.xlsx (+ .pdf)	Delivered
10	Logical Data Model	ER diagram and explanation	10_Logical_Data_Model.pdf, 10_ER_Diagram.pdf (+ .xlsx)	Delivered
11	KPI Catalogue	Spreadsheet, 30+ KPIs	11_KPI_Catalogue.xlsx (+ .pdf)	Delivered
12	Simulated Data Profiling	Workbook and findings	12_Simulated_Data_Profiling.xlsx, 12_Simulated_Data_Profiling.pdf (Users data only)	Delivered (Users only)
13	Data Quality Audit	Workbook and report	13_Data_Quality_Audit.xlsx, 13_Data_Quality_Audit.pdf (Users data only)	Delivered (Users only)
14	Dashboard Blueprint	Wireframes and specifications	14_Dashboard_Blueprint.pdf (+ .xlsx)	Delivered
15	Developer Analytics Handoff	Engineering specification	15_Developer_Analytics_Handoff.pdf (+ .xlsx)	Delivered
16	Data Pipeline Proposal	Architecture diagram and documentation	16_Data_Pipeline_Proposal.pdf, 16_Pipeline_Architecture_Diagram.pdf (+ .xlsx)	Delivered
17	Privacy and Governance Proposal	Document	17_Privacy_Governance_Proposal.pdf (+ .xlsx)	Delivered

#	Deliverable	Required form	File(s)	Status
18	Analytics Readiness Assessment	Scorecard and evidence	20_Analytics_Readiness_Assessment.pdf (+ .xlsx)	Delivered
19	Assumption and Verification Register	Spreadsheet	19_Assumption_Verification_Register.xlsx (+ .pdf)	Delivered
20	Implementation Backlog	Prioritised backlog	21_Implementation_Backlog.pdf (+ .xlsx)	Delivered
21	Final Consolidated Report	Professional document	Final_Consolidated_Report.pdf (this document)	Delivered
S1	Source and Verification Register (brief section 6)	Register	S1_Source_and_Verification_Register.xlsx (+ .pdf)	Supplementary
S2	Data Quality Control Framework (Work Package L)	Document	S2_Data_Quality_Control_Framework.pdf (+ .xlsx)	Supplementary
S3	Analytics Change Management (Work Package Q)	Document and log	S3_Analytics_Change_Management.pdf (+ .xlsx)	Supplementary

Core deliverables: 02 to 11, 14 to 17, 19, plus Executive_Summary and Final_Consolidated_Report.
Dataset limits: Deliverables 12 and 13 describe the Users entity only, because the other datasets were not supplied.
Numbering: Brief 18 is file 20 and brief 20 is file 21; the Source Register, Quality Framework and Change Management files are supplementary (S1, S2, S3).

4. Findings by work package

WP	Topic	Key content	File
A	Product Data Readiness Audit	20 product areas, user and system actions, 42-row coverage matrix, 15 analytics gaps	02, 03
B	User and Account Analytics	4 identifiers, 29 fields, lifecycle states, excluded personal data	04
C	Earning and Reward Analytics	Task lifecycle, 20 task fields, failure classes, reward events and transitions, third-party handling	05
D	Wallet and Withdrawal Analytics	Append-only ledger model (15 fields), withdrawal lifecycle (15 fields), 10 withdrawal KPIs, reconciliation checks	06
E	Referral and Engagement Analytics	4-stage referral funnel (13 fields), qualifying activity, engagement measures	07
F	Production Event Tracking Plan	42 events with the brief's 16 fields; naming standard; common payload	08
G	Data Dictionary	128 fields, 10 entities, sensitivity classes	09
H	Logical Data Model	Keys, 16 foreign keys, 20 relationships, ER diagram, integrity rules	10
I	KPI Catalogue	54 KPIs in 8 areas, none with a target	11
J	Simulated Data Profiling	Profile, relationship tests and KPI test on the Users data: 2 calculable, 5 partly, 47 not	12
K	Data Quality Audit	25 issues logged; 18 of 39 rules tested; 3 rules proposed	13
L	Data Quality Control Framework	39 rules, six layers, failure handling, quality metrics	S2
M	Analytics Dashboard Blueprint	7 dashboards, 56 visuals, wireframes	14
N	Developer Analytics Handoff	25 event requirements, 12 general requirements, 12 pre-conditions	15
O	Data Pipeline and Operating Model	13 components, two phases, run order, monitoring, failure handling, RACI	16
P	Data Access, Privacy and Governance	6 roles, 10 zones, 6 Restricted fields, do-not-expose list	17
Q	Analytics Change Management	9 triggers, impact matrix, versioning, change log	S3
R	Analytics Gap Assessment	10 areas, readiness scale, 31 gaps	18
S	Assumption and Verification Register	50 unknowns with owners and closing evidence	19
-	Final Implementation Backlog	43 items, waves, critical path	20

Findings from the supplied dataset

- The file holds the Users entity only: a clean sheet (1,000 rows) and an audit sheet (1,008 rows). Of the 54 KPIs, 2 are fully calculable, 5 partly and 47 not.
- Its own summary sheet matches a recomputation for all 8 metrics.
- The audit logged 25 issues. The audit sheet is the clean sheet plus 8 repeated rows and 12 changed cells; the 'clean' sheet itself breaks rules (negative derived days, logins after the period end, sessions without a login, marketing opt-in without consent).
- Stored derived fields (days_since_registration, activity flags) rely on an undocumented reference time and contradict other fields in places; the design choice to derive activity from events is supported.
- The file lacks a referrer link, so referral conversion cannot be measured; it has fields our dictionary lacks (email verification, consent, device) that are worth adding, and personal-looking fields our dictionary deliberately excludes.

5. Definition of done

The brief says the task is complete only when every item below is met. All items are now met, with two qualifications: profiling and the audit cover the Users entity only, because the other datasets were not supplied, and three items are met only as a proposal because nothing has been confirmed with the VELOOP team.

Item	Status	Evidence
Product data areas have been mapped	Met	02, 03
Major analytical events have been defined	Met	08
User analytics requirements are documented	Met	04
Reward analytics requirements are documented	Met	05
Wallet and withdrawal analytics are documented	Met	06
Referral analytics are documented	Met	07
Event naming is standardised	Met	08
Data dictionary is complete	Met as proposal	09: complete as a proposal; no field confirmed
Logical data model is complete	Met as proposal	10: complete as a proposal; not compared with real tables
KPIs have reproducible definitions	Met	11
Simulated data has been profiled	Met (Users only)	12: the supplied file holds the Users entity only (A046)
Data quality problems have been identified	Met (Users only)	13: problems in the Users audit sheet
Data quality controls have been proposed	Met	S2
Dashboard requirements are documented	Met	14
Engineering requirements are specific and testable	Met	15
Data pipeline is documented	Met	16
Governance and access requirements are documented	Met	17
Unknowns are explicitly recorded	Met	19
Analytics gaps are prioritised	Met	03, 18
Final implementation backlog exists	Met	20
All simulated data is clearly labelled	Met	Deliverables 12 and 13 carry the label SIMULATED DATA - CREATED FOR ANALYTICAL TESTING; other example values are marked illustrative
No fictional result is represented as an actual VELOOP result	Met	Results appear only in Deliverables 12 and 13, labelled as simulated
Final documents are internally consistent	Met as proposal	Cross-checks in section 7; documents not yet reviewed by the VELOOP team

6. Compliance with the integrity rules and the evaluation framework

Non-negotiable rule	How it was followed
1. Never fabricate production data	No production data was created. Example values are illustrative (for example usr_example_0001).
2. Never represent simulated data as actual VELOOP statistics	Simulated data appears only in Deliverables 12 and 13, labelled as simulated; counts describe the file, not VELOOP.
3. Never create fake company performance claims from the dataset	Figures appear only in Deliverables 12 and 13 and describe the simulated file, labelled as such. KPIs have formulas only; wireframes are labelled 'no data shown'.
4. Never invent confidential security mechanisms	None invented. The privacy proposal states it does not describe VELOOP's security controls; reasons are categories only.
5. Never request passwords, API keys or payment credentials	None requested. Section 6 of the privacy proposal lists what must never be exposed.
6. Never claim a proposed event already exists	Every event is Proposed or To Be Verified; Implementation status is 'To be confirmed by Engineering'.
7. Never hide uncertainty	50 unknowns registered; readiness levels stated honestly; the Users-only limit of the dataset work is stated.
8. Never convert an assumption into a fact	Nothing about VELOOP's product is marked Confirmed (two items about the supplied file are). Unrelated search results about similarly named products were not used.
9. Never add unnecessary personal-data fields	None exist; excluded data is listed.
10. Never optimise for appearance instead of accuracy	No targets, no results, readiness shown honestly.
11. Every important conclusion is traceable	KPIs, gaps and backlog items cite sources and IDs; a reference check found no invalid IDs.

Non-negotiable rule	How it was followed
12. Simulated data must be replaced by governed production data, not mixed	Separate zones, rule DQ039, a labelling rule and a cutover rule.

Evaluation category	Where it is addressed
Business and product value (25%)	Backlog, KPI Catalogue and dashboards tie every item to a decision; priorities carry reasons
Analytical design (20%)	Event, model and KPI design; four activity-versus-outcome distinctions; reproducible formulas
Data integrity and trust (20%)	Status labels, assumption register, no Confirmed items, honest readiness, simulated data kept apart
Engineering usefulness (15%)	Developer handoff with payloads, validation, deduplication keys and positive and negative acceptance tests
Data quality and governance (10%)	39 rules across six layers; quarantine; access model; change process
Documentation and presentation (10%)	Consistent structure, summaries, IDs, mapping and version table

7. Consistency checks

- The coverage matrix and the event tracking plan list the same 42 events, with the same sources, statuses and priorities.
- The event plan's KPI dependency column is generated from the KPI Catalogue's required-events column; 'Completion Rate' was split into Task Completion Rate and Withdrawal Completion Rate (change log CL-001).
- Every KPI appears on at least one dashboard (checked by formula).
- Every readiness gap and every open assumption maps to at least one backlog item (checked when the backlog was generated).
- A reference check across all PDFs found no invalid ID for assumptions, gaps, rules, KPIs, events, requirements or backlog items.
- The dataset's own summary sheet was recomputed from the data and all 8 metrics matched; every problem injected into the audit sheet was traced to exact rows.
- Three assumption citations that pointed at the wrong item (account statuses, pending reward state, country geography) were corrected and the affected documents regenerated.
- File numbers used inside documents (for example in the change impact matrix) are file numbers; the mapping in section 3 gives the brief's numbering.

8. Risks, limitations and next steps

Risk	Why it matters	Addressed by
Unverified foundations	The design rests on 50 unknowns; only 2 are confirmed (both about the supplied file) and 5 are Critical risk	Assumption Register; BL-30
Wallet integrity	Duplicate credits, or balances edited in place, would make wallet and withdrawal figures untrustworthy (A005, A023, A029)	BL-02, BL-04, BL-03
KPIs that change meaning	'Active user' and 'activation' are undecided, so core KPIs could shift once decided (A024, A010)	BL-07, BL-08
Mixed units	Adding amounts in different reward units would give false totals (A002)	BL-09; rule DQ035
Events for features that do not exist	Wasted effort and empty dashboards (A008)	BL-11
Simulated data mistaken for results	Fictional data used as if it were VELOOP's would mislead (brief rules 2 and 12)	Rule DQ039; BL-20
Privacy and legal unknowns	Retention, deletion and consent depend on where users are (A016, A036)	BL-29
Over-engineering	Infrastructure bought before it is needed adds cost and failure points	Two-phase pipeline with scale triggers
Rules and KPIs untested beyond Users	Only the Users data was supplied, so 21 rules and most KPIs could not be tested (A046)	BL-12

Limitations

- The supplied dataset covers the Users entity only, so relationship tests beyond Users, most KPI tests and several quality rules could not be run.
- No official VELOOP documentation was available; no product fact is verified.
- All statuses stay Proposed or To Be Verified until Engineering and Product confirm them.
- Legal points are outlines that need Legal review.

Next steps

- Provide the remaining datasets (task activity, rewards, wallet, withdrawals, referrals, feature events), or confirm they do not exist yet, so profiling, the KPI test and the audit can be re-run.
- Ask Engineering to confirm the Critical unknowns first (A002, A003, A005, A023, A029).
- Ask Product for the quick decisions: 'active', activation, units, feature list.
- Review the document set with Product and Engineering and move statuses from Proposed to Confirmed with evidence.
- Start the Wave 1 items in the backlog.

9. References

All deliverables listed in section 3 are the references for this report. Cross-references use these identifiers: A### (assumptions), G## and RG-## (gaps), DQ### (quality rules), KPI IDs such as E05 or D07, AE-### (event requirements), AG-## (general requirements), AP-## (pre-conditions), BL-## (backlog items), CL-### (change log).